

PROFESSIONAL SUMMARY

- Experienced Healthcare Data Engineer with 3+ years of hands-on experience in designing, implementing, and maintaining secure, scalable, and automated ETL and ELT pipelines for healthcare data exchange, analytics, and compliance.
- Specialized in processing complex healthcare EDI X12 transactions including 837 (Professional, Institutional, Dental), 835, 834, 270, and 271 across payer platforms such as Facets, QNXT, Amisys, and HealthEdge.
- Proficient in extracting, transforming, and loading clinical, claims, pharmacy, and eligibility data from EMR/EHR platforms including Epic, Cerner, AthenaHealth, and Salesforce Health Cloud, using HL7, FHIR, and custom API integration.
- Skilled in designing real-time and batch ingestion pipelines using Apache Kafka, Apache Spark, AWS Glue, Azure Data Factory, and Databricks, supporting use cases such as claim adjudication, eligibility verification, and encounter tracking.
- Experienced with Snowflake, Azure Synapse, and Amazon Redshift for data warehousing, analytics, data lake design, and development of HIPAA-compliant storage and compute environments.
- Strong command of SQL, Python, Scala, and Bash for data wrangling, data validation, schema transformation, and orchestrated workflow execution.
- Built modular transformation layers using dbt (data build tool), enabling version-controlled deployment of data models, documentation, testing, and integration with metadata tools such as Collibra.
- Implemented complex data governance controls including PHI and PII masking, tokenization, row-level security, access control, encryption at rest and in transit, and audit trail logging.
- Integrated and automated regulatory reporting pipelines for CMS, NCQA, T-MSIS, RAF, and HEDIS, aligning with federal and state reporting formats and compliance protocols.
- Created automated ingestion and validation workflows using Apache NiFi, SSIS, and custom Python frameworks to parse inbound 837/835/834 files, validate X12 loops, and route errors to exception handling layers.
- Experience working with HL7 v2.x, HL7 FHIR R4, X12 5010, NCPDP, UB-04, and CMS encounter formats within payer and provider interoperability environments.
- Hands-on with healthcare domain tools including Mirth Connect, Postman, EDI analyzers, ICD-10/CPT validation engines, CMS NPPES/NPI Registry, and plan hierarchy databases.
- Engineered eligibility and enrollment data flows for Medicaid and commercial plans using 834 feeds with member matching, retroactive coverage reconciliation, and payer-plan mapping logic.
- Supported payer-provider data exchange workflows including claim status updates, remittance advice, encounter submissions, and scheduling integration through HL7 messages and RESTful APIs.
- Developed ETL frameworks for pharmacy claims, Rx eligibility, formulary mapping, and NDC crosswalk standardization to support Medicare Part D, MTM, and PBM analytics.
- Integrated data quality tools and frameworks into CI/CD pipelines using GitHub Actions, Jenkins, Confluence, and Jira, ensuring continuous delivery and monitoring of healthcare data pipelines.
- Delivered analytics-ready datasets and curated marts for actuarial, financial, clinical, operational, and population health dashboards using Power BI, Tableau, and Looker.
- Conducted root cause analysis for claim rejections, invalid encounters, and eligibility mismatches using rule engines, audit logs, and historical data comparisons in Databricks and Snowflake.

South Carolina Department of Health (SCDHHS) Columbia, SC
Data Engineer

June 2024 – Present

Led real-time 270/271 eligibility inquiry integrations between state-contracted clinics and Medicaid systems by using Azure Event Grid and ADF to ingest HL7 messages and route responses with up-to-date member coverage status.

- Built dynamic converters to transform HL7 eligibility inquiries into JSON objects and join against the Medicaid Master Eligibility File to verify active plan, MCO, and benefit tier.
- Designed EMR-to-claim cross-validation logic for HCBS and EPSDT services by comparing encounter details submitted via HL7 ADT and ORU messages against 837 claim batches.
- Enabled streaming ingestion of clinical visit records into Synapse from over 100 providers via SSIS-based pipelines integrated with secure SFTP polling agents.
- Implemented masking logic in Synapse for behavioral health and SUD service claims per 42 CFR Part 2, including scoped views for authorized mental health reviewers only.
- Created encounter validation dashboards in Power BI, allowing Medicaid analysts to review pending, rejected, and completed encounter statuses alongside provider submission patterns.
- Authored SQL-based logic to flag retroactive eligibility mismatches between the 834 enrollment feed and HL7 registration records, automatically tagging incomplete coverage scenarios.
- Partnered with the CMS compliance team to build extract pipelines for T-MSIS submission, formatting encounter data to federal XML templates with metadata tagging.
- Integrated provider rosters from NPPES registry files to validate NPI, specialty, and service location attributes on all inbound encounter records.
- Built a daily eligibility sync routine that joined multiple data sources (834, 270/271, MMIS) to update Power BI dashboards with live member status for downstream care teams.
- Participated in user story review sessions with policy and IT teams to convert Medicaid business rules into eligibility validation logic within ADF.
- Designed appointment-level tracking tables for prenatal visit timeliness, EPSDT screening, and service code bundling validation, improving audit readiness.
- Delivered schema diagrams and documentation for all HL7 and 837 intake zones, helping train junior analysts on data flow structure and business logic.
- Ensured all integrations met HIPAA and CMS technical standards, with PII/PHI masking, audit logging, and secure transfer protocols.

Tech Stack: Azure Data Factory (ADF), Azure Synapse Analytics, Azure Event Grid, Azure Logic Apps, SSIS, T-SQL, SQL Server, HL7 v2.x, X12 (270/271, 834), Power BI, GitHub, Secure SFTP, JSON-to-HL7 Converters

**Tech Mahindra – Hyderabad, Telangana
Data Engineer**

Nov 2020 – Dec 2022

Built real-time ingestion pipelines using Apache NiFi and Mirth Connect to capture HL7 and FHIR-based resources (Patient, Encounter, Observation) from Epic and AthenaHealth EMRs, streaming data into Delta Lake tables within Databricks.

- Designed Spark-based reconciliation logic to compare real-time encounter data against adjudicated claims from the Clover CMS pipeline, surfacing mismatches in diagnosis, NPI attribution, and encounter-to-claim alignment.
- Modeled RAF and HCC scoring layers in dbt to quantify member risk stratification and code capture accuracy, integrating CMS weight models and clinical input from PCPs.
- Developed provider-facing Tableau dashboards displaying coding gaps, HCC persistence flags, risk opportunities by condition group, and STARS performance metrics.
- Integrated Part D Rx claims with EMR medication orders to flag unfilled prescriptions, dual therapy risks, and medication adherence gaps tied to Star Ratings.
- Authored logic to link ICD and SNOMED terminologies from clinical systems with submitted diagnosis codes on claims, ensuring coding consistency across data domains.
- Worked closely with the Compliance and RADV audit teams to extract EMR-claim match failures, maintain full lineage, and trace each submission for CMS reviews.
- Built FHIR-to-claim mapping logic for encounter matching, maintaining record-level audit logs with correlation IDs and ingestion timestamps.

- Designed Snowflake Secure Views and masking policies to protect PHI/PII across multiple consumer roles, including care coordination, analytics, and audit staff.
- Created reusable Spark job templates and modular NiFi flows to support expansion into additional clinical systems as Clover scaled its direct EHR integrations.
- Supported downstream analytics by maintaining curated condition-level timelines that reflected the full clinical and billing history of each member across all touchpoints.
- Led internal QA testing for EMR-to-claims reconciliation pipelines, implementing dbt test cases and monitoring logic to detect schema drifts, duplicate claims, and dropped diagnoses.
- Delivered documentation on all mappings, ETL logic, and exception handling workflows through GitHub and Confluence for engineering, compliance, and operations teams.

Tech Stack: Databricks (Spark, Delta Lake), Apache NiFi, Snowflake, dbt, PySpark, SQL, Mirth Connect, HL7 FHIR (R4), Tableau, GitHub, QNXT, ICD/SNOMED Mapping, Secure Views & Role-Based Masking, Confluence

Education

- **Masters of Science in Computer Science** - University of Illinois Springfield, Springfield, Illinois
- **Bachelors in Computer Science and Engineering** - MLR Institute of Technology, Affiliated with JNTUH, Hyderabad, Telangana